High-Dimensional Harmonic Dimension Comparison

Adaptive Riccati Transmission, Noncommuting Legs,
and Exact Tuning Certificates

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Abstract

The September 25 source constructs a nonnegatively Ricci-curved counterexample to integer-degree Euclidean harmonic dimension comparison by combining Berger-link spectra, finite simultaneous matrix control, and exactly tuned long radial periods. We separate the finite dimension-sixteen spectral certificate from the analytic construction of a word, a complete metric, and globally controlled harmonic functions. The source already contains integrated matrix damping, a scalar baseline, one-parameter-derivative estimates, and exact contraction tuning. Our additions are a history-independent adaptive baseline improving the pulse-to-adaptive-ideal error from the source's fixed-ideal order j^{-2} to j^{-6}, a noncommuting positive-leg factorization with a sharp Loewner bound, and explicit residual, determinant-normalization, and computed-Jacobian certificates. The adaptive ideal itself remains $O(j^{-2})$ from the fixed control word. All error remainders and positivity margins are retained. Exact rational fixtures and an explicit leakage example clarify why approximate transmission is not the global growth theorem. Contribution assessment: C3/5, proved source-relative analytic refinements and conditional certification tools, not a new global counterexample, historical priority claim, or blanket proof certification.

1 Problem, inherited theorem, and domains

For a complete pointed manifold, let

$$\mathcal{H}_k(M) = \{u \in C^\infty(M, \mathbb{R}) : \Delta_g u = 0,\ |u(x)| \le C_u(1 + d_g(o, x))^k\}.$$

For integer $k \ge 0$, in dimension $n = m + 1$,

$$\dim \mathcal{H}_k(\mathbb{R}^{m+1}) = \binom{m+k}{m} + \binom{m+k-1}{m}. \tag{1.1}$$

The second binomial is interpreted as zero when $k = 0$. The source [1] asserts that for some even $n \ge 8$ and integer $k \ge 2$ there is a smooth complete metric on $\mathbb{R}^n$, Euclidean near the origin, with $\mathrm{Ric} \ge 0$, asymptotic volume ratio in $(0, 1)$, and more than (1.1) harmonic functions of growth at

most k. Its appendix gives a finite spectral witness at $n = 16, k = 50000$. Neither that arithmetic nor the tests in this report alone produce the required finite control word or infinite sequence of exactly tuned periods.

Throughout the analytic additions there are only finitely many fixed harmonic blocks $2 \le l \le L$. Constants may depend on these blocks, the fixed compact parameter ball, and the fixed finite pulse family. They do not assert bounds uniform as L, the word length, or the control dimension tend to infinity. "Uniform over histories" means over every sequence of earlier controls in that fixed compact ball. A current-parameter derivative holds those earlier controls fixed.

The new results are Theorems 4.2, 5.1, 6.1, and 7.2, with Proposition 7.1. The source geometry, simultaneous Lie generation, finite spectral surplus, exact-cycle growth argument, and the distinction from locally conformally flat comparison theorems remain inherited. The noncommuting-leg result is an ODE extension: it does not assert that an arbitrary noncommuting angular path is realizable by a metric with the source's Ricci bounds.

2 Source mechanism and exact spectral selection

2.1 Berger links and the real multiplicities

Write $m = 2s - 1$, $s \ge 4$, and let V_l be the real round degree-l spherical harmonics on S^m, with

$$N_l = \binom{m+l}{m} - \binom{m+l-2}{m}, \qquad B_l = l(l+m-1).$$

For an orthogonal complex structure J, let $\xi_J(x) = Jx$, $\eta_J = \xi_J^\flat$ for the round metric, and D_J be differentiation along ξ_J. The source uses both orientation classes of J. Its links and positive angular Laplacian are

$$G(a, q, J) = a^2 q^{-1/m}\{g_0 + (q-1)\eta_J^2\}, \tag{2.1}$$

$$A_l(a, q, J) = a^{-2} q^{1/m}\{B_l I + (q^{-1} - 1)(-D_J^2)\}. \tag{2.2}$$

Their volume density is a^m times round volume. The two Ricci eigenvalues, with an index raised by G, are

$$a^{-2} q^{1/m}\{m - 1 - 2(q-1)\}, \qquad a^{-2} q^{1/m}(m-1)q.$$

Thus the strict link margin is precisely

$$a^2 < q^{1/m} \min\{1 - 2(q-1)/(m-1), q\}. \tag{2.3}$$

These are inherited identities, not a new curvature construction. In a Hopf weight decomposition, charge $2b - l$ has complex multiplicity

$$h_{l,b} = \frac{l+s-1}{s-1}\binom{b+s-2}{s-2}\binom{l-b+s-2}{s-2}. \tag{2.4}$$

It follows by subtracting the bidegree $(b-1, l-b-1)$ polynomials from bidegree $(b, l-b)$ polynomials. Pairing conjugate charges $b, l-b$ gives the real multiplicity $2h_{l,b}$ unless $2b = l$, when it is $h_{l,b}$. There is no additional doubling.

Put $a_*^2 = q_*^{1/m}\alpha_*^2$. The ordered positive growing roots of the dwell equation are

$$d_{l,i}(d_{l,i}+m-1) = \lambda_{l,i}, \quad \lambda_{l,b} = \alpha_*^{-2}\{B_l - (1-q_*^{-1})(2b-l)^2\}. \tag{2.5}$$

The selected number M_l is a prefix of this ordered real list, with mean $\bar{d}_l < k$, not necessarily a prefix all of whose members are less than k. A cut through a repeated eigenspace is permitted: a real orthonormal eigenbasis can be chosen inside it.

2.2 Asymptotic selection versus a finite certificate

The source first proves the existence of suitable data. Its beta-binomial weights converge to $X \sim \mathrm{Beta}(s-1, s-1)$, so $(2X-1)^2 \sim \mathrm{Beta}(1/2, (m-1)/2)$. If $y_m(v)$ is the upper-tail quantile and $J_m = \int_0^1 y_m(v)\log(1/v)\,dv$, its Gaussian-limit argument gives $(m-1)J_m > 2$ for sufficiently large odd m. Choosing $2/(m-1) < c < J_m$, $q_* = 1+\varepsilon$, and $\alpha_*^2 = 1 - c\varepsilon$, the source's prefix-count integral has first variation $m(J_m - c)/2 > 0$. Its layer-cake/Fatou argument then produces finite k, L, M_l with

$$\sum_{l=2}^{L} M_l > \sum_{l=0}^{k} N_l, \qquad \bar{d}_l < k \quad (M_l > 0). \tag{2.6}$$

This is a proof route, not the finite arithmetic check below. The cutoff smallness and strict link margin must be fixed before the control family; increasing the initial period index cannot retroactively change this spectral selection.

For the explicit witness take

$$m = 15, \quad s = 8, \quad k = 50000, \quad q_* = 101/100, \quad \alpha_*^2 = 99853/100000.$$

The horizontal margin is $29/700000 > 0$, and $101 \cdot 99853^{15} < 100 \cdot 100000^{15}$ proves $a_* < 1$. Let $Q_0 = 10^8$, $h_0 = 14$, and for each (l, b) set

$$\mathrm{num} = 100000\{101l(l+14) - (2b-l)^2\}, \quad \mathrm{den} = 101 \cdot 99853,$$

$$S = \mathrm{isqrt}\left(\left\lfloor Q_0^2(h_0^2\mathrm{den} + 4\mathrm{num})/\mathrm{den}\right\rfloor\right) + 1, \qquad u = \frac{S - h_0Q_0}{2Q_0} > d_{l,b}. \tag{2.7}$$

The added one guarantees strictness even for an exact square. Accept a whole multiplicity only if its accumulated upper sum is strictly below k times the accumulated dimension. If a block with integer root numerator $d_u > 2Q_0k$ fails, the admissible partial multiplicity is

$$\max\left\{0, \min\left\{\mathrm{mult}, \left\lfloor \frac{2Q_0k\,\mathrm{count} - \mathrm{upper_sum} - 1}{d_u - 2Q_0k} \right\rfloor\right\}\right\}.$$

All degrees through 49963 pass the sufficient full-degree test; degrees 49964 through 50212 are checked individually. The author's separate exact-integer replay obtains 16 full and 233 partial degrees in that latter range and the following totals:

$$\begin{aligned}
\mathrm{selected} &= 467856050444743403870888112576578170107146873664862179341111, \\
\mathrm{Euclidean} &= 467797093491463622392661265386095055118554921349866866362 51, \\
\mathrm{surplus} &= 589569532797814782268471904831149885919523149953129 7860.
\end{aligned}$$

The integers certify (2.6) at the specified parameters, not the geometric realization. A reproducible read/test ledger and hashes accompany this report.

2.3 What the simultaneous control theorem supplies

The source proves that the tuples $X_J = (\kappa_l(D_J^2)_{\mathrm{tf}})_{l=2}^L$ generate $\bigoplus_{l=2}^L \mathfrak{sl}(V_l)$, where

$$p_\infty = m-1, \quad \theta_l^2 + p_\infty \theta_l = a_*^{-2} B_l, \quad \kappa_l = \frac{a_*^{-2}}{p_\infty + 2\theta_l} > 0.$$

This nontrivial algebraic theorem is inherited, with its complete source proof read, rather than inferred from numerical ranks. Its actual mechanism is important. Sign choices of the complex structure supply diagonal weight operators. Joint adjoint interpolation separates source weights for shifts supported on at least three coordinates; only extreme weights are asserted one-dimensional. Explicit two-token split/merge entries join the extreme weights, including the odd-degree path using a fourth plane. The resulting isotropic rank-one powers and their orthogonal orbit span the harmonic representation, as proved using the Fischer decomposition and divisibility by the quadratic form. Their brackets give the skew matrices, and conjugation gives all trace-free symmetric matrices. The real algebra is therefore $\mathfrak{sl}(V_l)$. Distinct N_l exclude identifications between the simple factors.

Consequently there is a finite word $\mathcal{W}(h)$ with $\mathcal{W}(0) = \Pi$, $D\mathcal{W}(0)$ an isomorphism, where Π_l is a determinant-one signed cycle on the first M_l axes and identity on their complement. To pass from generation to this assertion, span the algebra by finitely many conjugates of the generators, use the inverse function theorem on their product at the identity, and use connectedness of real SL_N to reach Π by a finite word. This proves existence, not a manageable word length for the enormous finite witness. Smooth disjoint bumps $\beta_\nu \Subset (0,1)$, normalized to integral one, realize the signed flow coefficients $c_\nu(h)$.

3 Geometry, histories, and inherited exact transmission

3.1 Why the control choices preserve the geometry

The source fixes a nondecreasing smooth cutoff χ, zero for $t \le 2$ and one for $t \ge 4$, and sets

$$b(t) = a'(t)/a(t) = -\mu\chi(t)(1+t)^{-5/4}, \quad a(t) = \exp\left(\int_2^t b(w)\,dw\right), \quad a(t) \to a_*.$$

The fixed smallness condition gives $b \ge -1/4$. For $t \ge 4$,

$$\log(a(t)/a_*) = 4\mu(1+t)^{-1/4}, \quad b' = \frac{5\mu}{4}(1+t)^{-9/4}, \quad b'' = -\frac{45\mu}{16}(1+t)^{-13/4}. \tag{3.1}$$

Let $T_j = j^6, L_j = j^7, t_1 = 10$, and $t_{j+1} = t_j + L_j + 4T_j$. Each period consists, in order, of a pulse of length T_j, upward ramp of length T_j, dwell of length L_j, downward ramp of length T_j, and round rest of length T_j. On the pulse,

$$q = 1 + T_j^{-1} \sum_\nu c_\nu(h_j)\beta_\nu((t-t_j)/T_j),$$

and its active J_ν is constant on each bump. Labels change only on open round gaps. The other four pieces use one fixed J_0, the ramps join 1 and q_*, and the dwell has $q = q_*$.

For $g = dr^2 + r^2 G(a(\log r), q(\log r), J(\log r))$, put $B = bI + H$, where $H = (\dot{q}/2q)(P_J - I/m)$ and P_J is the Hopf projection. The shape operator is $r^{-1}(I + B)$, and the source proves

$$r^2 \operatorname{Ric}_{rr} = -m(b + \dot{b} + b^2) - \operatorname{tr}(H^2), \tag{3.2}$$

$$r^2 \operatorname{Ric}^{\#}_{\tan} = \operatorname{Ric}^{\#}_{\gamma} - \dot{B} - [m(1 + b) - 1](I + B), \tag{3.3}$$

$$\operatorname{tr}(H^2) = \frac{m - 1}{4m}(\dot{q}/q)^2. \tag{3.4}$$

The mixed component vanishes: the normalized Hopf field is a constant-length Killing field and hence geodesic and divergence-free, so $\operatorname{div}_\gamma P_J = 0$. In (3.3), the normal and tangential H^2 terms cancel; inserting an extra negative H^2 would not be the stated identity.

Here $t_j \sim j^8/8$. The positive radial term in (3.2) is bounded below by a constant times j^{-10}, whereas $\operatorname{tr} H^2 = O(j^{-12})$ on ramps and is smaller on pulses. The strict link margin (2.3) on the compact interval $[1, q_*]$, together with the strict margin at $(a_*, 1)$, handles both ramps and signed pulses, including $q < 1$. All remaining terms of (3.3) tend uniformly to their round-cone values. Before the first pulse, $f(r) = ra(\log r)$ satisfies $0 < f' \le 1, f'' \le 0$, giving nonnegative sectional curvature there. Smooth flat junctions, no finite accumulation of periods, and an exactly Euclidean origin yield a smooth metric. Every path from the origin has length at least its radial variation, and radial paths attain it. Thus $d_g(0, x) = r$; the centered closed balls are compact, which proves completeness. These arguments choose one starting index for every admitted control history.

3.2 Center-regular Riccati equation and the source pulse bound

The angular volume density is independent of the angular point. For each fixed V_l, the harmonic equation is therefore

$$Y'' + pY' - A_l Y = 0, \qquad p = m - 1 + mb, \qquad P' = A_l - pP - P^2, \quad P = lI \ (t \le 2). \tag{3.5}$$

There are constants $0 < \lambda_- \le \lambda_+$ with $\lambda_- I \preceq A_l \preceq \lambda_+ I$ uniformly, and $3m/4 - 1 \le p \le m - 1$. Choose $0 < c_P < C_P$ so that $\lambda_- - (m - 1)c_P - c_P^2 > 0$ and $\lambda_+ - (3m/4 - 1)C_P - C_P^2 < 0$, enclosing the initial l. At a first extremal eigenvalue contact these strict inequalities point inward. They prove global symmetric bounds $c_P I \preceq P \preceq C_P I$ and prevent finite-time blowup. The fundamental matrix of $Y' = PY$ has positive determinant and solves (3.5). Near the origin the other scalar solution is $r^{-(l+m-1)}$, which smoothness excludes. Thus this first-order system parametrizes exactly all center-regular solutions, with arbitrary values at any fixed positive radius.

For symmetric two-sided coefficients, $E' = F - \mathsf{L}E - E\mathsf{R}$ and $\lambda_{\min}\mathsf{L} + \lambda_{\min}\mathsf{R} \ge \gamma > 0$ imply

$$\|E(t)\|_F \le e^{-\gamma(t-u)} \|E(u)\|_F + \int_u^t e^{-\gamma(t-s)} \|F(s)\|_F \, ds, \quad \int_u^v \|E\|_F \le \frac{\|E(u)\|_F + \int_u^v \|F\|_F}{\gamma}. \tag{3.6}$$

Indeed, take the Frobenius inner product with E, divide the resulting scalar differential inequality by its norm (regularizing at zero), and integrate. No commutation is used. On the preceding round rest this gives $P(t_j) - \theta_l I = O(j^{-2})$, uniformly over histories, because the round forcing differs from its limiting constant by $O(j^{-2})$.

The source then uses a fresh exact scalar baseline $v' = a^{-2}B_l - pv - v^2, v(t_j) = \theta_l$, retaining $\varphi = \int_{t_j}^{t_j + T_j} v$ exactly. It sets

$$R_l(\tau, h) = \kappa_l \sum_\nu c_\nu(h)\beta_\nu(\tau)(B_l I/m + D^2_{J_\nu}), \qquad Q = vI + R_l/T_j.$$

Taylor expansion of (2.2) gives $\rho = A_l - pQ - Q^2 - Q' = O_{C_h^1}(\eta_j/T_j)$, where $\eta_j = j^{-2} + T_j^{-1}$. With $E = P - Q$, the exact ordered identities are

$$E' = \rho - (pI + P)E - EQ, \tag{3.7}$$

$$E'_h = \rho_h - (pI + P)E_h - E_hP - Q_hE - EQ_h. \tag{3.8}$$

The second identity holds with earlier controls fixed and $p_h = 0$. Equation (3.6) gives integrated C_h^1 error $O(\eta_j)$, not $O(T_j\eta_j)$. Variation of constants then gives

$$e^{-\varphi}\mathcal{B}_{l,j} = H_l(1,h) + O_{C_h^1}(\eta_j), \quad H_{l,\tau} = R_lH_l, \quad H_l(0) = I, \quad P_e = \theta_lI + O_{C_h^1}(\eta_j).$$

The integrated damping, scalar baseline, and current-parameter estimate in this paragraph are already proved in the source.

3.3 The whole leg and exact tuning, not approximate cycling

On the source's entire fixed-label leg, let U, V solve (3.5) with initial data $(I, 0), (0, I)$. In the fixed eigenbasis they are diagonal. At the leg endpoint set $D = U + \theta_lV$ and $K = D^{-1}V$. Scalar positivity gives $0 \preceq K \preceq \theta_l^{-1}I$, independent of the enormous condition number of D. The exact identity is

$$D^{-1}\mathcal{T}_{l,j} = [I + K(P_e - \theta_lI)]\mathcal{B}_{l,j}. \tag{3.9}$$

The bracket is on the left. The reference D, K is independent of the current and previous controls. The source proves

$$\log(D_{l,j})_{ii} = L_jd_{l,i} + O(1 + L_jj^{-2} + T_j).$$

Removing determinants by $\mathcal{N}_N(C) = (\det C)^{-1/N}C$ gives $\Psi_j(h) = \Phi(h) + O_{C_h^1}(\eta_j)$, where $\Phi = (\mathcal{N}_{N_l}(H_l(1,h)))_l = \mathcal{W}(h)$.

In a fixed local coordinate chart at Π, write $f = \zeta\Phi$, $f_j = \zeta\Psi_j$, and $A_0 = Df(0)$. Choose a small ball so that $\|I - A_0^{-1}Df\| \le 1/4$; a uniform large starting index ensures $\|A_0^{-1}(Df_j - Df)\| \le 1/4$ and $\|A_0^{-1}f_j(0)\| \le r/2$. The map $h \mapsto h - A_0^{-1}f_j(h)$ is a self-map contraction on that ball. Inductively, after fixing the actual previous history, choose its exact fixed point h_j. One threshold works for all histories. Thus

$$\mathcal{T}_{l,j} = s_{l,j}D_{l,j}\Pi_l, \qquad s_{l,j} > 0, \quad \log s_{l,j} = O(T_j). \tag{3.10}$$

This exact contraction argument is inherited, not an added result.

4 An adaptive history-independent radial baseline

Lemma 4.1 (Global round baseline)**.** *For the fixed source data there is a scalar function, independent of all controls and histories,*

$$v_l' = a^{-2}B_l - pv_l - v_l^2, \qquad v_l = l \quad (t \le 2). \tag{4.1}$$

It lies between fixed positive constants and satisfies, as $t \to \infty$,

$$v_l - \theta_l = O(t^{-1/4}), \quad v_l' = O(t^{-5/4}), \quad w_l(t) := \frac{a(t)^{-2}}{p(t) + 2v_l(t)} = \kappa_l + O(t^{-1/4}), \quad w_l' = O(t^{-5/4}).$$

All constants can be chosen for the fixed finite set of l.

Proof. The scalar versions of the strict barriers for (3.5) give positivity, boundedness, and global continuation. Subtracting the limiting root equation gives

$$(v_l - \theta_l)' = a^{-2}B_l - p\theta_l - \theta_l^2 - (p + v_l + \theta_l)(v_l - \theta_l).$$

The forcing is $O(t^{-1/4})$, and the damping has a positive lower bound. For a convolution against an exponentially decaying kernel, split $[4, t]$ at $t/2$: the first part is exponentially small and the second is bounded by a constant times the forcing order at t. This proves the first estimate. Differentiating (4.1), with $u = v_l'$, gives exactly

$$u' = (a^{-2}B_l)' - p'v_l - (p + 2v_l)u.$$

Here $(a^{-2})' = -2ba^{-2} = O(t^{-5/4})$ and $p' = mb' = O(t^{-9/4})$, by (3.1). The same convolution argument proves the estimate for u. The quotient rule, the strict positive denominator, and these bounds give both claims about w_l. Differentiation is legitimate since a, p are smooth and the scalar solution is smooth. □

Theorem 4.2 (Adaptive pulse and full-period refinement)**.** *Use the source's fixed pulse family and all sufficiently late periods, and hold earlier controls fixed when differentiating. On period j put $T = T_j$, $t = t_j + T\tau$, and define*

$$R_{l,j}(t, h) = w_l(t) \sum_{\nu} c_\nu(h)\beta_\nu(\tau)(B_l I/m + D_{J_\nu}^2), \tag{4.2}$$

$$\partial_\tau H_{l,j}(\tau, h) = R_{l,j}(t_j + T\tau, h)H_{l,j}(\tau, h), \qquad H_{l,j}(0, h) = I.$$

Let $\phi_{l,j} = \int_{t_j}^{t_j+T} v_l(t)\,dt$, $\theta_{l,j} = v_l(t_j + T)$, and define the full-leg reference $D_{l,j}^{\mathrm{ad}} = U + \theta_{l,j}V$. There are positive constants C, c, uniform over all admitted histories, such that

$$\left\| e^{-\phi_{l,j}}\mathcal{B}_{l,j} - H_{l,j}(1, \cdot) \right\|_{C_h^1} \le Cj^{-6}, \tag{4.3}$$

$$\|P_e - \theta_{l,j}I\|_{C_h^1} \le Ce^{-cT_j}, \tag{4.4}$$

$$\left\| e^{-\phi_{l,j}}(D_{l,j}^{\mathrm{ad}})^{-1}\mathcal{T}_{l,j} - H_{l,j}(1, \cdot) \right\|_{C_h^1} \le Cj^{-6}. \tag{4.5}$$

The adaptive normalized ideal $\Phi_j^{\mathrm{ad}} = (\mathcal{N}_{N_l}(H_{l,j}(1, \cdot)))_l$ differs from the fixed word Φ by $O_{C_h^1}(j^{-2})$. On a fixed sufficiently small coordinate ball there is a history-independent ideal control $h_j^{\mathrm{ad}} = O(j^{-2})$ hitting Π exactly. The actual adaptive-normalized map has a unique control on that ball and its control satisfies

$$\left\| h_j - h_j^{\mathrm{ad}} \right\| \le Cj^{-6}. \tag{4.6}$$

These are sharper comparisons to an adaptive reference, not an $O(j^{-6})$ comparison to the original fixed word.

Proof. On a round interval $E_0 = P - v_l I$ satisfies the homogeneous identity

$$E_0' = -(pI + P)E_0 - E_0 v_l I.$$

Positive barriers give a fixed damping rate. At the start of the last round rest its norm is uniformly bounded, so at t_j it is at most $Ce^{-cT_{j-1}}$. Before the first pulse, uniqueness gives $P = v_l I$ identically. Denote this initial error by ϵ_j, taking $\epsilon_{j_0} = 0$. Its derivative in the current parameter is zero.

Take $Q = v_l I + R_{l,j}/T$. Taylor expansion at the actual $a(t)$, not at a_*, gives

$$A_l = a^{-2} B_l I + (p + 2v_l) R_{l,j}/T + \mathcal{E}_A, \qquad \|\mathcal{E}_A\|_{C^1_h} \le C T^{-2}.$$

Since the fixed bumps have bounded derivatives, $\|\partial_t R_{l,j}\|_{C^1_h} \le C(T^{-1} + j^{-10})$, by Lemma 4.1 and $t \asymp j^8$. Hence the residual is exactly

$$\rho = \mathcal{E}_A - R_{l,j}^2/T^2 - (\partial_t R_{l,j})/T, \qquad \|\rho\|_{C^1_h} \le C(T^{-2} + j^{-10}/T). \tag{4.7}$$

For large j, Q has a uniform positive lower bound. Applying (3.7), (3.8), and integrated damping yields

$$\int_{t_j}^{t_j+T} \|P - Q\|_{C^1_h}\, dt \le C\{\epsilon_j + T^{-1} + j^{-10}\} = O(j^{-6}).$$

In the derivative estimate the extra forcing has norm at most $C\|P - Q\|/T$; its integrated contribution is bounded by the preceding value estimate divided by T. After extracting the scalar $e^{\int v_l}$, the actual and ideal equations have coefficients $R_{l,j}/T + (P - Q)$ and $R_{l,j}/T$. Their coefficient norms and first parameter derivatives have uniformly bounded integrals. Duhamel's formula and its derivative prove (4.3).

There is a fixed $\sigma > 0$ such that every bump vanishes on $[1 - \sigma, 1]$. On this final gap $Q = v_l I$ and the equation for $P - v_l I$ is homogeneous. Its value and parameter derivative at the start of the gap are uniformly bounded; the derivative equation has damping $pI + P$ on the left and P on the right. Both therefore decay exponentially over length σT, which proves (4.4). With $K = (D^{\mathrm{ad}}_{l,j})^{-1} V$, $0 \preceq K \preceq \theta_{l,j}^{-1} I$. This reference is independent of all controls. Substitution into the exact left factorization (3.9) proves (4.5).

Finally $w_l(t) - \kappa_l = O(j^{-2})$ on the pulse. The ideal coefficient and its current derivative thus differ from the fixed ones by $O(j^{-2})$ in rescaled time. Duhamel and determinant normalization on a compact subset of positive-determinant matrices give the asserted C^1 closeness. In fixed coordinates write f_j^{ad} for this ideal map and g_j for the actual map. Both have derivatives within a fixed contraction margin of $A_0 = Df(0)$, and $\left\|g_j - f_j^{\mathrm{ad}}\right\|_{C^1} \le C j^{-6}$. The self-map contraction argument gives their respective unique zeros, with the ideal zero $O(j^{-2})$ from zero. If $\left\|I - A_0^{-1} D g_j\right\| \le q < 1$, then comparison of the two fixed-point equations gives

$$(1 - q)\left\|h_j - h_j^{\mathrm{ad}}\right\| \le \left\|A_0^{-1}\right\| \left\|g_j(h_j^{\mathrm{ad}}) - f_j^{\mathrm{ad}}(h_j^{\mathrm{ad}})\right\|.$$

This proves (4.6), uniformly over histories. □

Remark 4.3. The reference change does not change dwell exponents. Entrywise, $D^{\mathrm{ad}}_{l,j} = D_{l,j}[I + K_{l,j}(\theta_{l,j} - \theta_l)]$ in the source diagonal basis, with $\theta_{l,j} - \theta_l = O(j^{-2})$. Thus its diagonal logarithms change by $O(j^{-2})$, and the inherited cycle-growth proof still applies to controls constructed with this reference. This is an alternative quantitative realization of the same source mechanism, not a new spectral surplus or dimension range.

5 A noncommuting positive-leg extension

Theorem 5.1 (Sharp matrix entrance bound)**.** *On a finite interval $[u, v]$, let p be real continuous and $A = A^{\mathsf{T}} \succeq 0$ continuous. Piecewise continuous coefficients with the usual continuously matched*

Cauchy data are also allowed. Let U, V solve $Y'' + pY' - AY = 0$ with data $(I, 0)$, $(0, I)$ at u. For a fixed symmetric positive definite matrix Θ, put $D = U + V\Theta$. Then U, D are invertible with positive determinant and

$$K(t) := D(t)^{-1}V(t) = K(t)^{\mathsf{T}}, \quad K(u) = 0, \qquad 0 \prec K(t) \prec \Theta^{-1} \quad (t > u). \tag{5.1}$$

No timewise commutation of $A(t)$, or commutation with Θ, is assumed. For any entrance derivative matrix P_e and incoming value map $\mathcal{B}$, the full value map satisfies exactly

$$D^{-1}(U + VP_e)\mathcal{B} = [I + K(P_e - \Theta)]\mathcal{B}. \tag{5.2}$$

The non-strict upper constant Θ^{-1} is sharp in this class.

Proof. Let $w(t) = \exp(\int_u^t p)$. If $U(t_0)x = 0$, put $y(t) = U(t)x$. Integration by parts, using $(wy')' = wAy$, gives

$$0 = [y^{\mathsf{T}} w y']_u^{t_0} = \int_u^{t_0} w\{|y'|^2 + y^{\mathsf{T}} A y\}\, dt.$$

Both terms are nonnegative. Thus $y' = 0$, and its terminal zero forces $x = 0$. This proves invertibility of U at every positive time; continuity gives $\det U > 0$. The self and mixed Wronskians give

$$U^{\mathsf{T}}U' - U'^{\mathsf{T}}U = 0, \qquad U^{\mathsf{T}}V' - U'^{\mathsf{T}}V = w^{-1}I.$$

Indeed, each Wronskian derivative is $-p$ times that Wronskian, since A is symmetric, and their initial values are $0, I$. The first identity makes $U'U^{-1}$ symmetric. Consequently

$$L(t) := U(t)^{-1}V(t) = \int_u^t w(s)^{-1}U(s)^{-1}U(s)^{-\mathsf{T}}\, ds. \tag{5.3}$$

Thus $L(t) \succ 0$ for $t > u$. Since $D = U(I + L\Theta)$, its determinant is positive: $L\Theta$ is similar to a positive definite matrix. Moreover

$$K = (I + L\Theta)^{-1}L = (L^{-1} + \Theta)^{-1}.$$

This proves symmetry and the strict Loewner bounds by inversion of positive definite matrices. The endpoint u follows by continuity. Expanding $U + VP_e = D + V(P_e - \Theta)$ proves the ordered identity. For sharpness take $A = 0, p = 0$ on an interval of length T. Then $K = (T^{-1}I + \Theta)^{-1} \to \Theta^{-1}$ as $T \to \infty$. The proof also holds interval by interval for piecewise continuous coefficients. □

Corollary 5.2 (No long-leg condition-number loss)**.** *Suppose the leg and Θ are independent of a parameter h. Let F be a scalar-normalized pulse and H its reference, and write $Z = P_e - \Theta$, $C = (I + KZ)F$, and $\kappa = \left\|\Theta^{-1}\right\|_{\mathrm{op}}$. With operator norms on multiplicative factors and Frobenius norms on errors,*

$$\|C - H\|_F \le \|F - H\|_F + \kappa \|Z\|_F \|F\|_{\mathrm{op}},$$
$$\|D_hC - D_hH\|_F \le \|D_hF - D_hH\|_F + \kappa\{\|D_hZ\|_F \|F\|_{\mathrm{op}} + \|Z\|_{\mathrm{op}} \|D_hF\|_F\}.$$

The derivative norms mean the supremum over unit parameter directions.

Proof. Subtract H from the exact expression for C, differentiate, and use $K_h = 0$ and Theorem 5.1. If the leg depends on h, the omitted term K_hZF must instead be retained. □

One also has $K' = w^{-1}D^{-1}D^{-\mathsf{T}} \succ 0$, from the same Wronskian with D, V; the initial self-Wronskian of D is zero because Θ is symmetric. This identity does not require D itself to be symmetric. In particular, replacing $D^{-1}V$ by VD^{-1} is unjustified.

6 Weighted residuals with one parameter derivative

Theorem 6.1 (Nonuniform damping ledger)**.** *Let P, Q be symmetric matrices on $[u, v]$, with the time and parameter regularity needed for (3.7)–(3.8); assume $p_h = 0$. Suppose, uniformly on a parameter ball,*

$$P(t) \succeq c(t)I, \quad Q(t) \succeq q_0(t)I, \quad \|\rho(t)\|_F \le r_0(t), \quad \|D_h\rho(t)\|_F \le r_1(t), \quad \|D_hQ(t)\|_{\mathrm{op}} \le q_1(t).$$

The bounds may be integrable functions and need not be constant. Set $\gamma_0 = p + c + q_0$, $\gamma_1 = p + 2c$, $G_i(t, s) = \exp(-\int_s^t \gamma_i)$, and

$$e(t) = G_0(t, u)e_0 + \int_u^t G_0(t, s)r_0(s)\, ds, \tag{6.1}$$

$$e_1(t) = G_1(t, u)e_{10} + \int_u^t G_1(t, s)\{r_1(s) + 2q_1(s)e(s)\}\, ds. \tag{6.2}$$

If $\|P(u) - Q(u)\|_F \le e_0$ and $\|D_h(P - Q)(u)\|_F \le e_{10}$, then $\|P - Q\|_F \le e$ and $\|D_h(P - Q)\|_F \le e_1$. Local negative damping is allowed; it is retained in the kernels.

For a parameter-independent scalar $v_0(t)$, set $S = Q - v_0I$, and let $F' = (S + E)F, G' = SG, F(u) = G(u) = I$, where $E = P - Q$. Define

$$a_0 = \int_u^v \|S\|_{\mathrm{op}}, \quad a_1 = \int_u^v \|D_hS\|_F, \quad b_0 = \int_u^v e, \quad b_1 = \int_u^v e_1.$$

At the endpoint,

$$\|F - G\|_F \le e^{a_0+b_0}b_0, \tag{6.3}$$

$$\|D_hF - D_hG\|_F \le e^{a_0+b_0}\{b_1 + b_0(a_1 + b_1)\}. \tag{6.4}$$

Proof. The Frobenius calculation proving (3.6) now uses the time-dependent integrating factor. It gives (6.1); applied to (3.8), with $\|Q_hE + EQ_h\|_F \le 2q_1e$, it gives (6.2), in every unit parameter direction. Regularizing the norm at zero and passing to the limit justifies the differential inequalities there. Integrable coefficient bounds follow by the same integral argument or approximation.

For the endpoint error, Duhamel gives

$$F(v, u) - G(v, u) = \int_u^v F(v, s)E(s)G(s, u)\, ds.$$

The product of the two propagator norm bounds is at most $e^{a_0+b_0}$, proving (6.3). Differentiate this exact integral. The term differentiating E contributes at most $e^{a_0+b_0}b_1$. For fixed s, differentiating $F(v, s)$ costs the integral of $\|S_h\| + \|E_h\|$ over $[s, v]$, while differentiating $G(s, u)$ costs the integral of $\|S_h\|$ over $[u, s]$. Their sum is bounded by $a_1 + b_1$, not twice a_1. Integration against $e(s)$ proves (6.4). □

The theorem extends the source's constant-damping estimate, not its existence theory. A negative interval can make these explicit bounds large. No contraction or long-time smallness follows without checking the actual kernel integrals. If earlier controls are differentiated, e_{10} cannot automatically be set to zero. Bounds for the physical full coefficient space are required; a finite set of test vectors without a frame estimate does not give these norms.

7 From computed endpoints to exact controls

7.1 Determinant normalization with an absolute singular margin

All norms in the next proposition are Frobenius norms except where marked. A derivative norm is the induced norm on the specified Euclidean parameter space. For several blocks use the orthogonal sum of their Frobenius spaces and the resulting Euclidean norm.

Proposition 7.1 (A positive-determinant certificate)**.** *Let $\widehat{C}, C$ be real N-by-N matrices with $\det \widehat{C} > 0$, $\sigma_{\min}(\widehat{C}) \geq s_0 > 0$, and $\left\|C - \widehat{C}\right\|_F \leq \delta_0 < s_0$. Put*

$$s = s_0 - \delta_0, \quad M = \left\|\widehat{C}\right\|_F + \delta_0, \qquad L_0 = s^{-1}\left(1 + \frac{M}{\sqrt{N}s}\right), \quad L_2 = s^{-1}\left(\frac{2M}{Ns^2} + \frac{2}{\sqrt{N}s}\right).$$

Then $\det C > 0$ and

$$\left\|\mathcal{N}_N(C) - \mathcal{N}_N(\widehat{C})\right\|_F \leq L_0\delta_0. \tag{7.1}$$

For parameter-dependent matrices, if these margins hold uniformly, $\left\|D_hC - D_h\widehat{C}\right\|_F \leq \delta_1$, and $\left\|D_h\widehat{C}\right\|_F \leq M_h$, then

$$\left\|D_h\mathcal{N}_N(C) - D_h\mathcal{N}_N(\widehat{C})\right\|_F \leq L_0\delta_1 + L_2\delta_0M_h. \tag{7.2}$$

Proof. Every point of the connecting segment has smallest singular value at least s, so its determinant cannot cross zero. If $d(C) = (\det C)^{-1/N}$ and $a_H = \operatorname{tr}(C^{-1}H)/N$, then

$$D\mathcal{N}_N(C)[H] = d(C)(H - a_HC), \qquad |a_H| \leq \|H\|_F/(\sqrt{N}s), \quad d(C) \leq s^{-1}.$$

The last bound follows from the product of the singular values. The second derivative is

$$D^2\mathcal{N}_N(C)[H,J] = d(C)\left[\left(a_Ha_J + \frac{\operatorname{tr}(C^{-1}JC^{-1}H)}{N}\right)C - a_HJ - a_JH\right].$$

Cauchy–Schwarz for the trace gives the claimed bounds L_0, L_2 on the first and second derivatives along the segment. Integrate the first derivative for (7.1). For the parameter bound split the difference into $D\mathcal{N}(C)(C_h - \widehat{C}_h)$ and $(D\mathcal{N}(C) - D\mathcal{N}(\widehat{C}))\widehat{C}_h$, and apply the second-derivative bound to the latter term. □

An absolute lower singular value is essential; a small relative condition number alone does not bound s^{-1}. The residual, rounding, ODE approximation, and any truncation errors must already be included in δ_0, δ_1. The proposition does not convert ordinary floating point output into a rigorous enclosure.

7.2 Explicit coordinates and an actual Jacobian

For each block delete the (N_l, N_l) entry from $\Pi_l^{-1}\mathcal{N}_{N_l}(C_l(h)) - I$, and stack the remaining entries to define $f(h) \in \mathbb{R}^d$, where $d = \sum_l (N_l^2 - 1)$. Then

$$f(h) = 0 \quad \Longleftrightarrow \quad \mathcal{N}_{N_l}(C_l(h)) = \Pi_l \quad \text{for every } l. \tag{7.3}$$

Indeed, only the last diagonal entry of $\Pi_l^{-1}\mathcal{N}(C_l)$ is then undetermined, and determinant one forces it to be one. Deletion has norm at most one; Π_l is orthogonal. Thus Proposition 7.1 bounds the coordinate errors without an unspecified chart constant. At the target this is an isomorphism on the trace-free tangent space, since trace determines the deleted entry.

Theorem 7.2 (A posteriori inverse certificate)**.** *Let the exact normalized endpoint map f of* (7.3) *be C^1 on a neighborhood of $\overline{B}(c,r) \subset \mathbb{R}^d$. Let $\widehat{f}(c)$ and a square computed matrix $\widehat{J}$ satisfy the certified bounds*

$$\left\|f(c) - \widehat{f}(c)\right\| \le \epsilon_f, \qquad \sup_{h\in\overline{B}(c,r)} \left\|Df(h) - \widehat{J}\right\|_{\mathrm{op}} \le \epsilon_J + Lr.$$

For a fixed nonsingular matrix B, suppose

$$q := \left\|I - B\widehat{J}\right\|_{\mathrm{op}} + \|B\|_{\mathrm{op}}\,(\epsilon_J + Lr) < 1, \tag{7.4}$$

$$\eta := \left\|B\widehat{f}(c)\right\| + \|B\|_{\mathrm{op}}\,\epsilon_f, \qquad \eta + qr \le r. \tag{7.5}$$

There is a unique exact control h_* in that closed ball, $f(h_*) = 0$, and $\|h_* - c\| \le \eta/(1-q)$. If $B = \widehat{J}^{-1}$ with a certified $\sigma_{\min}(\widehat{J}) \ge \sigma_0 > 0$, it suffices to use $q = (\epsilon_J + Lr)/\sigma_0$ and $\eta = (\left\|\widehat{f}(c)\right\| + \epsilon_f)/\sigma_0$.

Proof. The derivative of $T(h) = h - Bf(h)$ has norm at most q. Integrating the derivative on the convex ball makes T q-Lipschitz. Also $\|T(c) - c\| \le \eta$, so $\|T(h) - c\| \le \eta + qr \le r$. Completeness of the closed ball and the contraction theorem give a unique fixed point. Invertibility of B turns $Bf(h_*) = 0$ into $f(h_*) = 0$; a rectangular left inverse would not justify this inference. The same estimate at the fixed point gives the displacement bound. Finally $\left\|\widehat{J}^{-1}\right\| \le 1/\sigma_0$ proves the stated sufficient constants. Equation (7.3) supplies the exact simultaneous target, not only its projection. □

This theorem uses the actual computed parameter Jacobian and retained errors, whereas the source proves an asymptotic comparison to a fixed ideal derivative. The contraction principle itself is conventional. The source-specific addition is the chain of explicit physical endpoint, determinant, and Jacobian margins. No interval enclosures for the full enormous source word have been supplied here, so this is a proved conditional certificate rather than a reported numerical construction of the counterexample. Repeating it at every period requires uniform certified constants over all admitted histories, or a separately checked certificate at each actual finite history.

8 Exact cycles, leakage, and global growth

8.1 The source's final analytic implication

Assume the source spectral data, geometry, simultaneous word theorem, and exact tuning have been supplied. For one selected initial axis in V_l, (3.10) moves its value through the selected M_l-cycle. Signs do not affect its norm. At the end of J periods its logarithmic norm is

$$\sum_{j=j_0}^{J} \{\log s_{l,j} + j^7 d_{l,i(j)} + O(j^6 + j^5 + 1)\}.$$

The sequence $d_{l,i(j)} - \bar{d}_l$ is periodic with zero mean, and hence has bounded partial sums. Summation by parts against j^7 bounds its weighted sum by $O(J^7)$. Also $\sum_{j \le J} \log s_{l,j} = O(J^7)$, whereas $t_{J+1} = J^8/8 + O(J^7)$. Thus the logarithmic growth divided by t_{J+1} tends to $\bar{d}_l < k$.

At intermediate times the bound $\|P_l\| \le C_P$ costs at most a constant times the current period length $O(J^7)$, which is $o(t_J)$. The same upper growth bound therefore holds at all large radii, not merely at a chosen sequence. Fix $\bar{d}_l < \alpha_l < k$. Since the angular space is finite dimensional, its coefficient norm controls the pointwise angular norm, giving $|u| \le C r^{\alpha_l}$ outside a compact set. Smoothness controls the compact set. The functions are independent because their values at the first pulse are independent within each block, and different harmonic blocks are orthogonal there. Together with (2.6), this proves the inherited dimension conclusion. The argument works for the adaptive reference as well, by the remark following Theorem 4.2.

For completeness, the Euclidean comparator really is the polynomial dimension. Convolve an entire Euclidean harmonic function of growth $O((1 + |x|)^k)$ with a smooth radial unit-mass mollifier at scale R. The mean value property makes this convolution the original function. Differentiating the kernel gives, for $|\alpha| > k$, $|\partial^\alpha u(x)| \le C_x R^{k-|\alpha|} \to 0$. Thus u is a harmonic polynomial of degree at most k. The homogeneous dimensions telescope to (1.1).

8.2 A small residual cannot replace exact tuning

Take abstract two-dimensional value transfers with one selected axis, $\Pi = I$, and

$$D_j = \operatorname{diag}(e^{aL_j}, e^{bL_j}), \qquad a < k < b.$$

For any $\epsilon \neq 0$, let $\mathcal{T}_1 = D_1(I + \epsilon E_{21})$ and $\mathcal{T}_j = D_j$ for $j \ge 2$. The normalized errors are supported in one period and are arbitrarily small; the shear has determinant one. Nevertheless, with $S_J = \sum_{j \le J} L_j$,

$$\mathcal{T}_J \cdots \mathcal{T}_1 e_1 = (e^{aS_J}, \epsilon e^{bS_J})^{\mathsf{T}}, \qquad S_J^{-1} \log \|\mathcal{T}_J \cdots \mathcal{T}_1 e_1\| \to b.$$

Hence even a summable, indeed finitely supported, endpoint error does not preserve the prescribed selected-axis growth. This does not assert nonexistence of another slow initial subspace, nor does it claim that this abstract fixture is a Ricci-nonnegative metric. It identifies the exact implication that a residual-only replacement of the source's cycle argument would fail to establish.

8.3 A genuinely noncommuting exact fixture

For Theorem 5.1, take $p = 0$, successive intervals of length $\log 2$, and

$$A_1 = \operatorname{diag}(1, 4), \quad A_2 = O \operatorname{diag}(9, 16) O^{\mathsf{T}}, \quad O = \frac{1}{5}\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad \Theta = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}.$$

Here $[A_1, A_2] \neq 0$. If $R = A^{1/2}$, the exact segment propagator on value/derivative pairs is

$$\begin{pmatrix} \cosh(Rt) & R^{-1}\sinh(Rt) \\ R\sinh(Rt) & \cosh(Rt) \end{pmatrix}.$$

All entries at the specified durations are rational since $\cosh(q \log 2) = (2^q + 2^{-q})/2$ and $\sinh(q \log 2) = (2^q - 2^{-q})/2$ for integers q. Multiplying the second segment on the left gives exact rational U, V. The test script checks the two Wronskians, positivity of L, K, and $\Theta^{-1} - K$, the Gram formula

for K', and both incorrect matrix-order alternatives. Eight positive entrance matrices are tested, without a scalar-K degeneracy. These finite identities support transcription checks; the theorem for arbitrary coefficients is established by its written proof.

9 Primary literature and the scope of comparison

The source is an OpenAI Math Release preprint dated September 25, 2026, pinned here to repository commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The public repository README and specified primary records were checked on October 8, 2026. A repository release and an exact arithmetic script are not independent acceptance of the full proof; no peer-review status or comprehensive literature-validity verdict is asserted by this report.

Cai–Lai's current primary record is arXiv:2608.04553v2, dated August 16, 2026 (v1 August 5). Their Theorem A assumes completeness, noncompactness, $n \geq 3$, nonnegative Ricci curvature, and *local conformal flatness*; it proves the Euclidean dimension bound for every real degree $d \geq 0$, with rigidity at an integer equality degree at least one [2]. Its additional geometric assumption is essential to the comparison being made here.

The source already distinguishes this subclass. On a long nonround dwell the metric is conformal to $dt^2 + h$ with non-Einstein Berger link h. In total dimension $m + 1$,

$$W_{0a0b} = -\frac{1}{m-1}\left((\mathrm{Ric}_h)_{ab} - \frac{\mathrm{Scal}_h}{m} h_{ab}\right).$$

This is nonzero for its nonround link, so the constructed metric is not locally conformally flat. Likewise the round and nonround links give nonisometric tangent cones. Huang's Theorem 1.6 assumes a *unique* tangent cone at infinity as well as maximal volume growth [3]. Its proof takes a uniformly controlled finite-dimensional family to independent harmonic functions on that one cone. Xu's Theorem 3.4 requires the comparison exponent to lie outside the union of cone homogeneity spectra [4]; the source proves that this union contains a neighborhood of its selected integer. None of these distinctions is a contribution of this report.

Only the stated external passages were read: Cai–Lai's introduction, Theorem A, and the displayed radial zero-volume proof in Theorem 2.2; Huang's Theorem 1.6 and its concluding proof; and Xu's Theorem 3.4, its proof and displayed local estimate. Their deeper geometric-limit dependencies have not all been independently read or reproved. The local results of this report do not require those external comparison proofs as lemmas.

10 Private proof lineage, benefits, and limitations

10.1 The original passages actually used

The private originals were read locally, not uploaded or queried online. TR212 [5] separates local jet recurrence from global boundary closure. Its finite trace criterion uses full column rank plus range compatibility, and its continuous residual theorem assumes the combined interior/boundary operator is an isomorphism. Its resonant Dirichlet example shows why a local noncharacteristic equation does not alone imply global invertibility. In the present report this distinction becomes the separation between Riccati propagation and the actual square parameter inverse in Theorem 7.2.

The Wronskian energy test is a direct boundary uniqueness argument, not an invocation of a jet formula as an inverse theorem.

The complete original `stability.tex` chapter of Report 1 [6] proves the residual-ball implication $q < 1, \eta + qr \le r$, assuming a bounded inverse-type preconditioner. Its transfer statement retains full operator errors and compatible spaces. We use that logic with explicit endpoint coordinates and singular margins; renaming the contraction theorem is not counted as new mathematics.

The actual core-pattern-5 proof passages [7] retain physical metrics, a Neumann norm margin, eliminated-variable residuals in Schur recovery, and the absolute singular margin in finite-rank updates. For example, with an interior residual r_I, Schur reconstruction contributes $A_{BI}A_{II}^{-1}r_I$ to the boundary residual. Analogously, this report carries $P_e - \Theta$ through the full leg instead of discarding it when normalizing. The physical coefficient metric is the fixed round L^2 inner product in each V_l; all matrix norm and parameter-coordinate changes must be accounted for. No assertion about the validity of the whole private corpus follows from these selected proof reads.

10.2 Pros, cons, and next mathematical steps

The adaptive theorem removes the artificial round-tail forcing from the pulse-start discrepancy and identifies a much smaller actual-to-adaptive error. Its scalar functions are independent of every history, and its exact final round gap is a particularly effective entrance certificate. The noncommuting-leg theorem avoids long-leg condition numbers without assuming simultaneous diagonalization. The weighted ledger allows nonuniform damping, while the normalization and inverse tests explicitly connect numerical enclosures to an exact target. These are source-specific analytic gains with complete proofs.

There are substantial limitations. Constants depend on the fixed finite representation and word and have not been optimized. No full word is numerically constructed at the dimension-sixteen witness. The noncommuting-leg theorem does not replace the diagonal cycle counting mechanism or solve the geometric realization of arbitrary angular matrices. Small endpoint residuals are not substitutes for exactly invariant selected directions. The sharper adaptive error does not improve the $O(j^{-2})$ distance to the fixed word or the source's global dimension range. No unproved infinite-dimensional limit, omitted coefficient tail, or finite-test-to-global implication is used.

A useful next step is a certified, reduced-size source control word, with interval enclosures for its actual Jacobian, the scalar baseline, and its parameter-dependent pulse. The quantities in Theorem 7.2 specify what such a computation must deliver. A separate geometric problem is to characterize noncommuting legs whose moving links retain a suitable divergence and Ricci structure; only after solving it could Theorem 5.1 support a different global metric construction. Optimizing the fixed spectral witness requires a new count and all its geometric margins, not merely a smaller residual bound.

11 Read scope, tests, and contribution assessment

Every extracted source file was literally read: `paper.tex` (109 lines); `introduction.tex` (189), `links.tex` (182), `counting.tex` (350), `algebra.tex` (500), `geometry.tex` (321), `transmission.tex` (758), `conclusion.tex` (201), `finite-selection.tex` (94), and `bibliography.tex` (121); the period-figure TeX (27); the top-level README (27), verification README (17), and recorded count output (9). An initially truncated algebra retrieval was replaced by complete reads of lines

1–238 and 239–500. The figure code was read; no independent source-PDF visual audit is claimed. The verification Python script was absent from the extracted tree; its complete 79-line pinned public version was read separately. The author's replay is a separate implementation of the exact inequalities, with direct multiplicity spot checks and a retained per-degree integer slack.

The private reads, with one-based inclusive line ranges, were: TR212, 1–306, 444–630, 759–858 of 1053 lines; `stability.tex`, all 295 lines; its `foundations.tex` caller, 269–305 of 326; the Report-1 main file, 39–46 of 64 for metadata; and core-pattern-5, 35–119, 189–309, 364–531 of 925. Unread portions are not included in the review claim. Exact paths and complete SHA-256 hashes, public passage scopes, test outputs, and final deliverable hashes are recorded in the owned audit sidecars.

The test suite reports 607 named exact checks, including the noncommuting two-segment fixtures, differentiated residual and normalization identities, a rational inverse-certificate example, and all 249 individually selected degrees. Its integer replay also checks 2,490,502 strict square-root upper bounds. These are finite algebra and arithmetic, not interval ODE validation or a computer proof of the manifold theorem. Parent batch checks are separately identified by their relevant Riccati and full-leg names; the combined batch total is not attributed to this manuscript.

No verified source error was found in this bounded examination of the stated mechanisms. That conditional statement is not independent certification of every source theorem or every cited paper. The demonstrated additions justify C3: a quantitative adaptive refinement, a positive noncommuting ODE extension, and a conditional solver-to-exact-control certificate with explicit errors. The original counterexample, its geometry, and its finite selection remain source results. Applicability level L4 is not the contribution score; C4 or C5 is not claimed for the present analytic refinements.

References

[1] OpenAI, *A counterexample to integer-degree harmonic dimension comparison*, OpenAI Math Release preprint, September 25, 2026. Pinned repository source and verification record.

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[4] G. Xu, *Three circles theorems for harmonic functions*, arXiv:1601.02066v1, January 9, 2016. `https://arxiv.org/abs/1601.02066v1`.

[5] M. Subbarao, *Boundary-Value Ram Operators: Trace Closure, Panel Assembly, Green Operators, Stability Certificates, and Elliptic and Oscillatory Applications*, Technical Report 2026-08-05-0212, August 5, 2026. Confidential local original; selected proof passages specified in Section 11.

[6] M. Subbarao, *Ram Calculus for Linear and Nonlinear Integral and Partial Differential Equations: Report 1, Localization, Moment Representations, and Certified Finite Inversion*, September 18, 2026. Confidential local original, `stability.tex` and specified caller passages.

[7] M. Subbarao, *Adaptive Local–Global Graph Operators with Certified Updates and Reconstruction*, core-pattern-5 technical report, October 1, 2026. Confidential local original; selected metric, frame, Neumann, Schur, and update proofs.